

Contribution Journal — Lab 04

SingleEpoch | ITAI 1371 Module 04

Individual contribution statement

This lab was completed as an individual effort in association with the group SingleEpoch. There was no peer group to divide work with, so the tasks below cover the full scope of the assignment.

Environment and repo setup. The project scaffold (`.venv` , `requirements.txt` , `setup_gate.py` , tests, CI workflow) was set up, the local gate was brought to passing, and the repo was initialized and connected to the GitHub remote.

A notebook defect surfaced before work could start. Running the first code cell produced a `SyntaxError: unterminated string literal` . Tracing it back showed a notebook authoring defect: a `print()` string had been split across two entries in the notebook's underlying source data, which broke the string open when the cell ran. The same problem turned up in a second cell. Every other code cell was checked to confirm the defect was isolated to those two, and both were fixed. The full RCA is documented in the reflective journal.

Every cell was run and verified individually. For each of the six instructor-filled code cells and both experiment cells, the cell was run and its output checked against a specific expectation before moving on, rather than running everything at once and assuming it worked.

Experiment code was written for the two student sections. For Experiment 1 (Embarked countplot) and Experiment 2 (Fare boxplot), the plotting code was written by mirroring the pattern already used in the instructor cells, and the resulting plots were read to answer what they showed about survival.

Knowledge check and reflections reflect direct observation. The knowledge check answers and the reflective journal are based on observations from the plots that were run, not pulled from an outside source.

Data quality claims were checked against the source. Where a factual claim about the dataset appears (missing value counts for Age, Cabin, Embarked), it was verified against the live dataset rather than assumed.